

SHUO YIN

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EDUCATION

University of Toronto

BASc, Computer Engineering + Co-op

- GPA: 3.86

- Awards & Certifications: **AWS Certified Cloud Practitioner**, Wallberg Scholarship, Edward Rogers Scholarship

Toronto, ON

Expected Graduation: Apr. 2028

EXPERIENCE

Undergraduate ML Researcher

Apr. 2026 - Present

READY Lab

Toronto, ON

- Programmed software that learns from 3D CAD designs and suggests a designer's next step using PyTorch, Open Cascade, and ETL.
- Trained a PyTorch model on 200,000+ 3D CAD samples to predict a designer's next modeling step from real design sequences.
- Built an Open Cascade ETL pipeline that converted raw CAD files into clean training data, cutting preparation time by an estimated 70%.
- Used pytest to validate data, API responses, and model predictions, catching errors before they could distort benchmark results.
- Standardized Anaconda dependencies across macOS and Linux, reducing environment setup and troubleshooting time by about 40%.
- Presented weekly reviews across multiple labs to explain results, flag blockers, and turn researcher feedback into code changes.

Autonomous Systems Developer

Sep. 2025 - Apr. 2026

University of Toronto Formula SAE Racing

Toronto, ON

- Programmed Python software that checks how well 8 YOLO object-detection models find objects in 5,000+ camera images, with GitHub Actions rerunning tests after code changes.
- Built Python APIs and CLI commands to run identical YOLO tests, giving teammates a repeatable way to compare accuracy and speed.
- Used Pandas to clean camera data and combine test results into comparison tables, cutting manual processing by an estimated 60%.
- Configured GitHub Actions to rerun benchmarks after code changes, cutting manual testing by about 50% and catching failures early.
- Wrote setup, testing, and debugging guides so teammates could install, run, and troubleshoot the system without one-on-one help.

Software Developer

Jan. 2025 - Aug. 2025

University of Toronto Aerospace Team

Toronto, ON

- Programmed a drone navigation system that planned safe flight paths and connected them to the aircraft using C++, ROS, and PX4.
- Built a C++ pathfinding service in ROS that sent planned routes to PX4 so the drone could follow them during autonomous flights.
- Built routing APIs to reject bad flight settings before PX4 received them, preventing invalid requests from breaking the navigation system.
- Used Gazebo to test navigation and obstacle handling across 50+ simulated flights, finding software issues before real-drone tests.
- Used Docker to standardize the ROS environment across teammates, reducing setup time by 30%+ and avoiding local setup problems.

PROJECTS

CADence Scholar | *LangGraph, Pinecone, Gemini API, React*

[GitHub Link](#)

- Programmed an app that reads research papers, answers questions, and cites source pages using LangGraph, Pinecone, and Gemini.
- Used Gemini embeddings and Pinecone to match questions with relevant paper sections even when users used different wording, making source evidence easier to find and verify.
- Used LangGraph to find evidence, create answers, verify citations, and retry unsupported responses before showing them to users.
- Used React and PyMuPDF to process multiple PDFs and show page-level citations so users could quickly verify each answer's source.

Neurolense VR | *Python, FastAPI, Three.js, WebXR*

[GitHub Link](#)

- Programmed an MRI app that turns medical scans into interactive 3D models for browser and VR using FastAPI, Three.js, and WebXR.
- Built FastAPI endpoints to upload MRI scans and create 3D models in the background so the page stayed responsive during processing.
- Built a NumPy and nibabel ETL pipeline that converted raw MRI scans into sliceable 3D models so users could inspect different layers.
- Used Three.js and WebXR to display MRI models in browser and VR, letting users rotate, slice, and inspect scans from different angles.

SKILLS

Programming Languages: Python, Java, SQL, C/C++, JavaScript, TypeScript, Bash, HTML/CSS

Frameworks & Libraries: React, Node.js, Three.js, FastAPI, NumPy, Pandas, SQLAlchemy, BeautifulSoup4, Chakra UI, REST APIs

Databases & Cloud: PostgreSQL, MySQL, Firebase, Firestore, AWS, Oracle Cloud, Google Cloud Platform, NoSQL, Serverless Functions

Developer Tools: Git, GitHub, GitLab, Linux/Unix, Docker, Postman, PuTTY, VS Code, Jupyter Notebook, GitHub Actions, CI/CD